Mohammad Alaei

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Education

- **Master of Science** — University of Tabriz, Iran
  Computer Engineering – Computer Systems Architecture; ***GPA: 18.93/20, 1st Rank*** — *Sep 2018 – Sep 2021*
  Thesis: Deep Learning-Based Estimation of Frontal EEG-Correlated Signals from Facial Video
  Supervisor: Dr. Hadi S. Aghdasi
  Selected for University Exceptional Talent Group

- **Bachelor of Science** — University of Tabriz, Iran
  Computer Engineering – Computer Hardware Engineering; ***GPA: 17.11/20, 3rd Rank*** — *Sep 2014 – Jul 2018*
  Final Project: Design and Development of a Robotic Music-Based Attention Training System for Children with ADHD
  Supervisor: Dr. Hadi S. Aghdasi

Research Interests

- **Digital Health Technologies**
- **Human–AI Interaction**
- **Personalized Intelligent Systems**
- **Computational Modeling of Human Behavior**

Research Experience

- **Research Internship** — Remote
  National Elite Foundation of Iran and JobVision (Part-time) — *Oct 2020 - March 2021*
  - Investigated critical employer-demanded soft skills through research and analysis
  - Designed and evaluated quantification approaches using online cognitive games, assessing their reliability and effectiveness
  - Synthesized research findings into actionable recommendations for game-based assessment strategies, collaborating within a 6-member interdisciplinary research team

- **Software Developer (VR Developer)** — University of Tabriz
  National Elite Foundation of Iran (Part-time) — *Oct 2019 - Sep 2020*
  - Developed a research-driven VR educational platform for teaching 3D geometry, deployed in multiple schools and evaluated for learning impact
  - Implemented immersive VR functionalities using C#, Unity, and HTC Vive to boost 3D geometry learning
  - Collaborated in a 10-member multidisciplinary team, contributing to core VR system design and implementation
  - Designed user interfaces to support usability, consistency, and experimental clarity in educational settings

- **Research Assistant** — University of Tabriz
  Humanoid Robots and Cognitive Technology Research Lab (Part-time) — *Sep 2018 - Feb 2019*
  - Reviewed assistive technology literature for cognitive and functional impairments to guide future research
  - Synthesized and presented research findings at weekly lab meetings, informing discussions and project planning
  - Mentored students on projects and theses, providing guidance on experimental design and methodology

Publications & Preprints

- **Mohammad Alaei**, Shabnam Oskouei, Hadi S. Aghdasi: "Estimating Frontal EEG–Correlated Signals from Facial Video Using a Spatiotemporal 3D CNN: An Intra-Subject Study", Submitted to *Scientific Reports*, 2026.

Selected Research Projects

- **Multisensory Stimulus Experiment Platform**: Developed a Python-based platform for synchronized auditory and visual stimulus presentation in cognitive experiments, supporting configurable experimental protocols and precise response logging. Implemented multithreaded stimulus presentation and response collection with timestamped event logging for experimental data analysis.

- **Self-Balancing Two-Wheeled Robot**: Developed a two-wheeled self-balancing robot using an IMU with sensor fusion and PID control for real-time balance stabilization. Implemented and tuned embedded control algorithms using IMU-derived orientation feedback for stable operation.

Professional Experience

- **Research Engineer**
  Kavida *Mar 2025 – Present*
  - Identified platform requirements through user and market needs analysis and designed a goal-oriented collaboration platform
  - Designed and implemented the end-to-end platform architecture, including backend services, APIs, database schema, and frontend components
  - Designed an extensible recommendation framework based on LLM embeddings, semantic user representations, and similarity-based matching of users and collaboration opportunities
  - Defined data models and feature representations to support future experimentation with recommendation systems, embedding strategies, and user modeling approaches

  Technologies: Python, Django REST framework (DRF), WebSockets, PostgreSQL, Sentence Transformers, React.js, Git
- **Software Engineer**
  Herrmann Innovations *Sep 2023 – May 2025*
  - Engineered a real-time vehicle analysis platform using React.js and Django REST API, integrating IoT sensor data and computer vision algorithms for data acquisition and decision support
  - Designed and automated a Raspberry Pi–based streaming infrastructure enabling real-time data exchange and scalable integration of camera-based monitoring systems

  Technologies: Python, DRF, React.js, JavaScript, PostgreSQL, Real-Time Data Streaming, Raspberry Pi, IoT Systems Integration, Git
- **AI & Software Developer**
  Civil Aviation Organization of Iran *Feb 2022 – May 2023*
  - Developed optimization and decision-support algorithms (A* and Risk A*) for complex path planning and operational efficiency problems within a 10-member engineering team.
  - Conducted literature review of academic and industry research to inform algorithm design and validated solution performance
  - Developed a map-based simulation platform with custom front-end and visualization for testing and analyzing optimization algorithms
- **Research Software Engineer & Full-Stack Developer**
  Freelance *Mar 2018 - Present*
  - Designed and deployed full-stack web applications and real-time experimental platforms for scalable, research-grade systems with data collection and automation capabilities
  - Engineered algorithms and prototypes across robotics, cryptography, and human-centered computing, incorporating real-time processing, sensor integration, and system-level optimization
  - Developed integrated UI/UX, database, and hardware–software systems in multidisciplinary environments, ensuring data integrity, reproducibility, and robust system performance

Technical Skills

- **Programming Languages**: Python, JavaScript, C/C++, C#, SQL
- **ML & AI**: PyTorch, TensorFlow, Keras, Scikit-Learn
- **Web & Backend**: Django, Django REST Framework (DRF), REST APIs, React.js
- **Databases**: PostgreSQL, MySQL
- **Tools**: Git, Unity, Docker

Language Skills

- **English**: Academic Proficiency
- **Azerbaijani**: Mother Tongue
- **Persian**: Native

Certificates

- Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University *Nov 2024*
- The Interactive Track and the Course Project of Deep Learning, Neuromatch Academy *Jul 2023*

References

- **Prof. Hadi S. Aghdasi, Full Professor, Department of Electrical and Computer Engineering, University of Tabriz, Iran**: Email: aghdasi@tabrizu.ac.ir
- **Prof. Mina Zolfy Lighvan, Associate Professor, Department of Electrical and Computer Engineering, University of Tabriz, Iran**: Email: mzolfy@tabrizu.ac.ir